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TITLE: LAB 8: THE PROCESS OF INTERACTION DESIGN ACTIVITY

NAME PRODUCT: HYDROTIME WATERBOTTLE

a) DISCOVERING REQUIREMENTS FOR THE INTERACTIVE PRODUCT

- Interviews and Surveys: Conduct interviews and surveys with potential users to understand their needs and preferences for a timekeeping water bottle. This includes fitness enthusiasts, office workers, students, etc. Besides, include questions about the importance and frequency of having hot and cold water available.
- Contextual Inquiry: Observe users in their natural environment (gyms, offices, schools) to see how they use water bottles and timepieces. Furthermore, observe how users manage their hot and cold drinks, especially in different environments like gyms, offices, and outdoor settings.
- Competitive Analysis: Analyse existing smart bottles and smartwatches to identify gaps and opportunities for innovation. Analyse existing thermos bottles and smart temperature control bottles to identify strengths and weaknesses.

For functional requirements, we provide these requirements for our HydroTime WaterBottle:

- Timekeeping: Must display time and date clearly.
- Hydration Tracking: Measure and track water intake.
- Reminders: Provide reminders to drink water, exercise, take breaks, etc.
- Fitness Tracking: Record exercise habits like steps, distance, and calories burned.
- Voice Interaction: Include voice assistance for hands-free operation.
- Durability and Waterproofing: Ensure the device is durable and water-resistant.
- Temperature detector: A digital screen display temperature of the water either it's cold, warm or hot.

Non-functional Requirements:

- Battery Life: Long-lasting battery with efficient energy use.
- User Interface: Intuitive and easy-to-read display.
- Aesthetics: Stylish and appealing design.
- Ergonomics: Comfortable to hold and use.

b) DESIGNING ALTERNATIVES THAT MEET THOSE REQUIREMENTS.

Design Alternatives:

- Design A: A digital screen embedded in the bottle cap displaying time, hydration levels, and notifications.
- Design B: An analog watch face embedded in a band around the bottle with smart sensors inside the bottle for tracking.
- Design C: A removable smart sleeve that fits over the bottle, containing all electronic components and display.

Material Selection for HydroTime Water Bottle:

Material: Stainless Steel (Grade 304 or 316)

Reasons:

Durability: Highly resistant to impacts and corrosion, ensuring a long lifespan.

Insulation: Good thermal properties for maintaining hot or cold water temperatures.

Aesthetics: Sleek, modern look that can be finished in various colors and textures.

Safety: Non-reactive and safe for food and beverage storage.

Temperature Control System: Thermoelectric Modules with Vacuum Insulation Layer

c) **PROTOTYPING THE ALTERNATIVES DESIGNS SO THAT THEY CAN BE COMMUNICATED AND ASSESSED.**

Low-Fidelity Prototypes: Sketches



- Using cheap bottle and wrap the digital watch around the bottle.

d) **EVALUATING THE PRODUCT AND THE USER EXPERIENCE IT OFFERS THROUGHOUT THE PROCESS.**

Formative Evaluation:

- **Heuristic Evaluation:** Have experts review the product against usability heuristics to identify potential issues early.
- **User Feedback:** Continuously gather user feedback during prototype testing to inform design iterations.

Summative Evaluation:

- **Usability Testing:** Conduct extensive usability testing with a larger group of users to assess the overall user experience, efficiency, and satisfaction.
- **Performance Metrics:** Measure key performance indicators such as accuracy of hydration tracking, battery life, and ease of use.



HYDROTIME WATERBOTTLE